

Research Statement

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My pursuit of a career in scientific research is driven by a commitment to bridge the gap between data and real-world impact. In the realm of translational research, I focus on turning technical biomedical findings into tools that can guide clinical decision making and improve community health. At the same time, I develop and apply cutting-edge methods in functional data analysis, which allow us to capture and interpret information contained in rich, dynamic data sources such as electronic health records (EHR), imaging, and sensors. Through the exploration of both topics, I aim to build open-source research toolkits that are methodologically innovative, deeply connected to the challenges of modern biomedical science and are accessible to the wider community.

1 Translational statistics

My academic journey began as a research assistant at the Center for Innovative Design & Analysis (CIDA) at the University of Colorado Anschutz Medical Campus—a collaborative hub advancing healthcare research through statistics and data science. In this role, I participated in a diverse variety of interdisciplinary projects, working closely with medical and public health professionals across vast domains, such as pulmonary medicine, critical care, epidemiology, digital health, and mental health. This journey keeps extending as I graduated and started working as a postdoctoral fellow at the National Institute of Environmental Health Sciences (NIEHS), supporting scientists in neurobiology and toxicology with the analysis of lab data. As a statistician, I explore and analyze these datasets based on their specific features and investigators' needs, provide intuitive visualization for interpretation, reveal new discoveries and guide decision making. These experiences have made me realize the profoundness of statistical analysis in scientific investigations and thus raised my passion in the new and arising field of translational statistics.

Generally speaking, translational statistics refer to the analysis of data and the presentation of results that are designed to facilitate their use in practical settings. Consequently, its focus goes beyond traditional methodological development, but extends to what can be learned from the data and how these discoveries should be implemented in real life. While the stakeholders, such as scientists, doctors and patients, often do not have statistical background, it is crucial that they understand these results and can utilize them for their own benefits. It highlights the fact that, in a collaborative team, statisticians must not only be good analysts but also efficient communicators. Translation methods developed for intuitive and practical interpretation are in demand to bridge the gap of interdisciplinary communication. In biomedicine research, these methods are often referred to as "bench-to-bedside" approaches, while in environmental science and epidemiology they become "bench-to-community"[1].

Since joining NIEHS, I have made several innovative tools to facilitate the "bench-to-community" translation. In my collaboration with the National Center for Complementary and Integrative Health (NCCIH), I have developed a semi-automated, highly reproducible toolkit for the detection of toxicity. Based on the principles of Shape-Restricted Regression Splines (SHARP) procedure[6], a dose-response curve will undergo a series of four shape tests: increase, decrease,

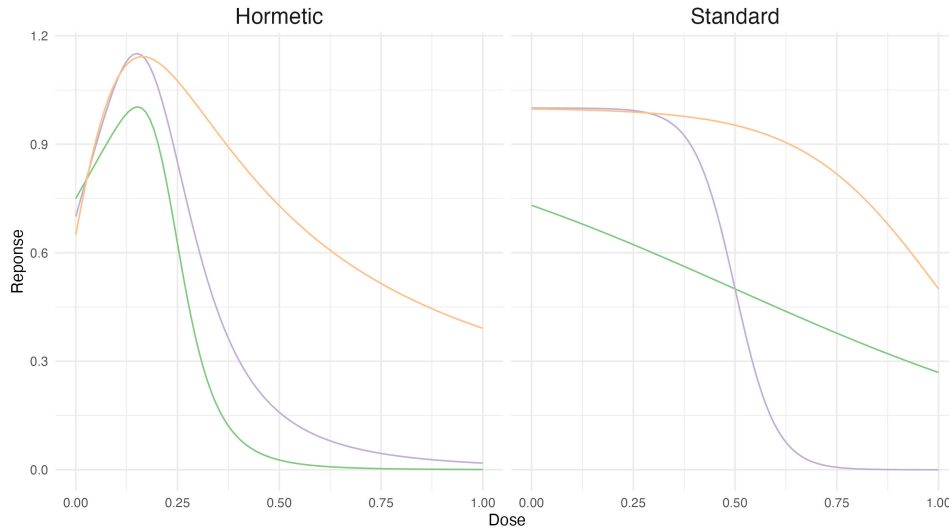


Figure 1: Simulated examples of hormetic dose-response curves and standard monotone dose-response curves

convex and concave. The resulting shape is matched to a type of toxicological pattern. Monotonically increasing or decreasing curves could reveal a typical adverse or beneficial effect of a chemical. However, decrease-concave or increase-convex curves reveal the possibility of hormesis, where a harmful substance can have beneficial effects at a very low dose. The difference between the two types of toxicology relationships is illustrated in Figure 1. To promote the interpretability of the shape detection results, especially the challenge of interpreting p-values, I have also created a visualization scheme that can present many dose-response curves on one integrative 2D scatterplot, and is capable of showing not only the shape detection results, but also the confidence level of these conclusions. As shown in Figure 2, each dose-response pair is represented by one point and their positions on the surface correspond to their shapes. Significant shapes are positioned along the edges out of significance boundaries ($p=0.05$) indicated by dashed lines, while insignificant shapes are positioned in the center within. The higher p-values are, the closer to the axis they will be, meaning the more indefinite the conclusions are. These tools have greatly improved the efficiency of toxicology screening because they help lab scientists read numerous dose-response curves all at once and offers an objective, rigorous standard to detect meaningful biological effects. This toolkit has been wrapped into an open-source R package and is in the process of being submitted for an NIH grant for translational science.

In addition, I am also interested in the translation of large complicated data, such as high-dimensional data, multivariate longitudinal data and spatial-temporal data. Since these datasets are often analyzed with efficient machine learning methods, translational statistics in this field is also referred to as explainable machine learning. I have proposed efficient, dynamic visualization methods of variable intercorrelation for both panel and longitudinal data, and created interactive tools for users without technical background. I am confident these tools and methods will greatly contribute to the scientific collaboration at Wake Forest University. On the other hand, these tools can also be utilized in classrooms to engage students with the interactive features and straightforward presentation.

2 Functional data analysis

Functional data analysis (FDA) is a young but rapidly growing topic, evolving in synchronization with hardware technology advances. Nowadays, portable sensors and wearable device are

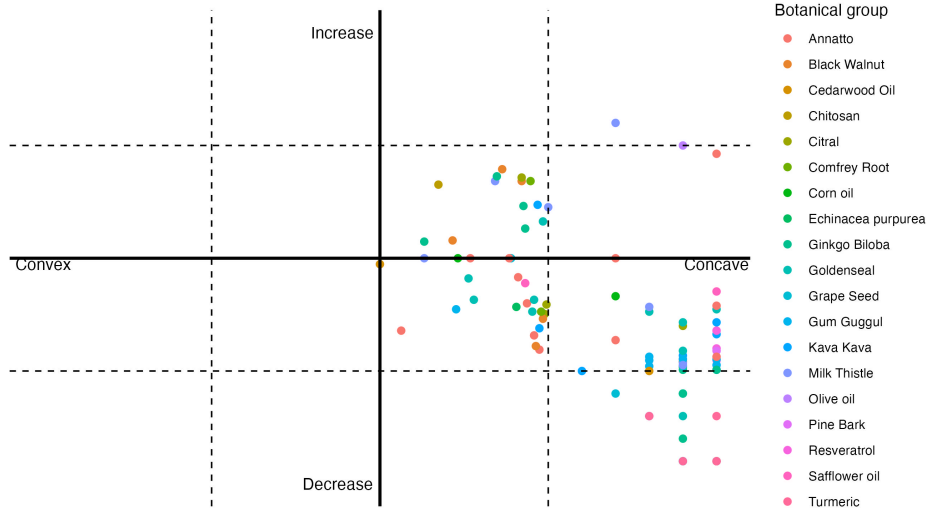


Figure 2: Scatterplot of the shape detection results of one response versus 66 chemicals. Dashed lines are the significance threshold corresponding to $p = 0.05$. Points placed in the center area have insignificant findings, while points out of the threshold lines showed significant shapes.

powerful enough to collect and store vast amount of data at very high-density continuously and simultaneously. For example, a single fitness tracker can collect activity, heart rate and sleep status at up to 80Hz, meaning 80 vectors per second[5]. One environmental sensors are capable of detecting the concentration of several dozens of pollutants at the same time, and doing so continuously for years. Such large scale and detailed data should have opened up potentials for many scientific investigations that were not previously feasible. Unfortunately, these datasets are often poorly utilized in reality, due to the lack of efficient, interpretable methods. Their size and complex structures impose severe computational burden beyond the capacity of conventional methods and devices. Deep learning models, though efficient enough, are very hard to interpret or communicate. Scientists and policy makers are reluctant to trust black-box analysis, especially on the subject of human health, making these models detached from practical use.

Functional data analysis was born out of these challenges. The set of methodologies essentially perceives high-density, complicated repeated measures as discrete realizations of an unobserved continuous process. Based on Karhunen-Loève theorem, such continuous processes can be decomposed into the linear combination of a set of orthogonal functions with proper scaling. Therefore, it can be estimated by a limited number of orthogonal basis functions, which significantly reduces the number of parameters to estimate. Based on this core assumption, many models have been developed, targeting a variety of statistical problem, including dimension reduction, prediction and inference. Recent work has also been expanding into more intricate topics such as causal inference and visualization. These methods can flexibly capture the complex shape of the curve, allowing for extraction of detailed information. Their application on physical activity data has revealed the effect of circadian rhythm on physical[4] and mental health[3], which is a great demonstration of the advantages of these methods over traditional approaches, such as summary statistics and mixed models.

I first learned about FDA through a doctorate-level elective course and decided to explore this field for my doctoral dissertation. Encouraged by my advisor, I joined an multi-institutional R01 project (#NS060910) on the analysis of longitudinal neuroimaging data. The high-density nature and spatial dependency of MRI images made it a good application for FDA methods. This experience has been a great opportunity to explore, learn and practice, and eventually inspired a few new ideas. I proposed a new dynamic prediction framework for high-density repeated measures based on fast Generalized Functional Principal Component Analysis (fGFPCA), predicting

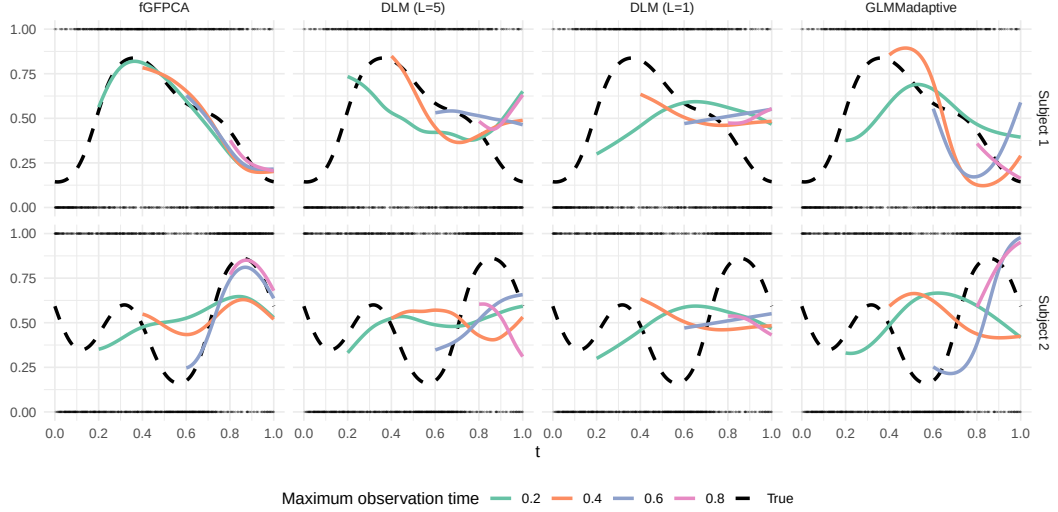


Figure 3: Simulated example of the dynamic prediction process using different methods. The black dashed line is the true unobserved underlying process. The colored lines are predicted tracks, given different observation interval.

the future trend using the historical records from the same individual. Through extensive simulation and a case study on the minute-level physical activity record from the CDC’s National Health and Nutrition Examination Survey[2], the proposed method showed great advantages in prediction accuracy by improving AUC by up to 30% compared to several other methods, including Distributed Lag Model (DLM) with different lags (L) and Generalized Linear Mixed Model with Adaptive Gaussian Quadrature (GLMMadaptive). In Figure 3, it is obvious that fGFPCA has achieved much greater performance as its predicted tracks (solid colored lines) are much closer to the truth (dashed black lines). As the observation period extends from 0.2 to 0.8, more data is collected, and the predictive performance of fGFPCA also improves with it. Among the four models listed, only fGFPCA shows this real-time improvement, which is a desired property of this type of predictive models. It is also efficient enough in terms of computation to be implemented on a portable device. Moreover, this method can be easily transferred to many other settings with minor model adjustments, such as predicting adverse events (e.g. cardiovascular attack) or incorporating covariates. It has broad practical applications for personalized real-time risk monitoring, which is an important goal of digital health research.

After joining NIEHS, my expertise in FDA has been applied to neurobiology research. I have been entrusted with Electroencephalogram (EEG) data collected from mice using implanted sensors to study the difference of sleep patterns between males and females, which steeps into the functional reference problem. In my future career, I wish to deepen my exploration of FDA methodologies and broaden their applications. One of such potential direction is data harmonization across different devices. Sensors from different companies can have very different hardware and preprocessing methods of raw data. These discrepancies have caused great inconsistency in measures and made it difficult to compare data across devices. I have been curious with the potential of a "harmonization model" that is able to learn and correct the differences between devices, similar to image registration. This project could be a great addition to the current research portfolio of the Department of Statistical Science. With the resources Wake Forest University can offer, I am excited to lead this extension into the world of digital health.

3 Future directions

As I transition to the next stage of my career, I am eager to establish myself as an independent researcher and leader in my field. I look forward to continuing interdisciplinary collaborations, particularly on projects that involve large, complex, and dynamic datasets. Through deeper engagement with domain experts, I aim to gain a more nuanced understanding of the scientific foundations and challenges associated with high-density, high-dimensional repeated measures. These insights will inspire new research questions on the translation of functional data analysis, which connects my two research interests above. I aim to develop this question into well-structured research projects and publications, as well as competitive grant proposals. Within the next three years, my goal is to secure independent research funding as a principal investigator, contributing not only to my professional growth but also to the advancement of the Department of Statistical Science.

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